COURSEDUP

REM UND APP ANALYZE EVA CRE

Ecology

Final Exam

INSTRUCTOR	UNIVERSITY	CODE
Sharon Patterson	**Virtual Earth University**	**GTR426**
FACULTY	**DEPARTMENT**	**WEEK**
Life Sciences	**Biology**	**16**
DISCIPLINE	**TYPE**	**LEVEL**
General	**Final Exam**	**Undergraduate**

THEORETICAL FRAMEWORK

- Foundations of Advanced Ecology
- Population Dynamics
- Community Interactions and Succession
- Biogeochemical Cycles
- Spatial Patterns and Fragmentation
- Restoration Ecology
- Co-evolution and Speciation
- Ecological Modeling and Data Analysis
- Ecosystem Services

Final Exam: Advanced Ecology and Contemporary Synthesis

Learning Objectives Assessed:

- Synthesize foundational principles of advanced ecology and methodological approaches from early coursework to understand basic ecological organization.
- Evaluate complex population dynamics and community interactions using advanced quantitative models.
- Analyze ecosystem-level processes, including biogeochemical cycles, energy flow, and landscape fragmentation.
- Assess the impacts of global climate change, conservation strategies, and habitat restoration initiatives.
- Critique contemporary ecological issues, evolutionary mechanisms, ecological economics, and advanced data analysis frameworks.

Duration: 120 minutes **Total:** 100 points **Type:** Final Exam

Powered by CoursedUp AI. Content may contain inaccuracies; please cross-check all outputs before use.

EXAM SCOPE & TOPIC WEIGHTING

The Final Exam for Ecology (GTR426) is fully cumulative, covering all core concepts and methodologies introduced from Week 1 through Week 14. To reflect the progressive mastery required in advanced coursework, the examination utilizes a recency-weighted design. While foundational early topics such as population dynamics and ecosystem ecology establish vital baseline knowledge, the assessment heavily emphasizes recent weeks. Specifically, Weeks 9 through 14 dominate the point distribution, accounting for over 60% of the total grade. Students must demonstrate higher-order cognitive skills, integrating historical ecological principles with emerging challenges in contemporary environmental science.

Coverage	Topic(s)	Weight	What to Focus On
Week 14	Contemporary Issues in Ecology: Emerging Challenges & Innovative Solutions	**14%**	Synthesize emerging environmental challenges with innovative, interdisciplinary solutions discussed in recent literature.
Week 13	Ecosystem Services & Ecological Economics	**13%**	Connect ecosystem services valuation methods directly to ecological economics and policy decision-making.
Week 12	Ecological Modeling & Data Analysis: Advanced Statistical Methods	**11%**	Practice interpreting advanced statistical models and ecological data sets to derive defensible conclusions.
Week 11	Evolutionary Ecology: Co-evolution, Adaptation, and Speciation	**10%**	Analyze co-evolutionary dynamics and speciation mechanisms within changing biotic and abiotic environments.
Week 10	Applied Ecology II: Urban Ecology & Human-Wildlife Conflict	**9%**	Examine how urban ecosystems function and evaluate successful mitigation strategies for human-wildlife conflict.
Week 9	Applied Ecology I: Restoration Ecology Principles and Practices	**8%**	Master the primary principles of ecological restoration, focusing on reference ecosystems and successional trajectories.
Week 7	Conservation Biology: Threats, Strategies, and Protected Areas Management	**7%**	Understand the quantitative frameworks used to design protected areas and prioritize conservation threats.
Week 6	Global Ecology & Climate Change Impacts	**6%**	Review global climate change feedback loops and their direct impacts on biogeographical distribution shifts.
Week 5	Landscape Ecology: Spatial Patterns, Fragmentation, and Connectivity	**5%**	Focus on how landscape fragmentation metrics directly influence species connectivity and metapopulation persistence.
Week 4	Ecosystem Ecology: Energy Flow, Nutrient Cycling, and Biogeochemical Cycles	**5%**	Prioritize tracing energy flow pathways and nutrient cycling efficiencies across various biome types.
Week 3	Community Ecology: Interactions, Diversity, and Succession	**4%**	
Week 2	Population Dynamics: Advanced Models & Conservation Applications	**4%**	
Week 1	Foundations of Advanced Ecology & Methodological Approaches	**4%**	

EXAM FORMAT & RULES

The examination is divided into three fixed sections designed to test a comprehensive range of cognitive skills, maintaining a target difficulty of 9/10. Part A consists of 10 Multiple Choice questions worth a total of 20 points, focusing on rapid recall, conceptual definitions, and foundational principles across all weeks. Part B comprises 5 Short Answer questions worth 40 points, requiring targeted problem-solving, data interpretation, and mechanical application of ecological models. Part C features 2 Essay and Integrative questions worth 40 points, demanding

deep critical synthesis, high-level theoretical application, and multi-week connections. The section weighting directly mirrors the topical coverage weighting, emphasizing rigorous analytical and integrative capabilities.

Section	Question Type	#	Points	Bloom Focus
Part A	Multiple Choice	10	20	Understand/Apply
Part B	Short Answer	5	40	Apply/Analyze
Part C	Essay / Integrative	2	40	Analyze/Evaluate/Create

Part A: Comprehensive multiple-choice items spanning the whole course.

Part B: Short-answer prompts across the full syllabus.

Part C: Extended integrative questions connecting concepts from different parts of the course.

EXAM RULES

- Arrive at the testing venue at least 15 minutes prior to the official start time with valid student identification.
- Bring only approved writing implements, a non-programmable scientific calculator, and a blank scratch paper provided by proctors.
- All electronic devices, including smartphones, smartwatches, and laptops, must be completely powered off and stowed away.
- The use of unauthorized textbooks, notes, digital resources, or generative artificial intelligence tools is strictly prohibited.
- Read all prompt instructions carefully and manage your time effectively across Parts A, B, and C.
- Show all mathematical workings and intermediate steps clearly in the provided space for partial credit consideration.
- Submit your complete examination packet directly to the instructor or proctor before departing the room.

EXAM QUESTION SET

This set matches the structure in the Exam Format table above. Week and difficulty tags show how the topic weighting maps to individual questions.

Part A: Multiple Choice

Choose the single best answer.

1. When evaluating the ecological impacts of novel synthetic pollutants like engineered nanomaterials in aquatic systems, which emerging synthesis approach best integrates multi-trophic bioaccumulation models with real-time sensor telemetry? [Week 14 | Hard]

A. Static steady-state mass balance modeling without dynamic exposure pathways

B. Dynamic probabilistic life-cycle assessment coupled with physiologically based toxicokinetic-toxicodynamic (PBTK-TD) networks

C. Empirical bivariate correlation of single-species mortality against instantaneous chemical concentration

D. Purely qualitative risk matrix mapping based on expert elicitation without field validation

2. In ecological economics, when valuing ecosystem services that exhibit non-linear threshold responses (such as lake eutrophication), which valuation technique is most appropriate to capture the true economic loss associated with regime shifts? [Week 13 | Medium]

A. Travel cost method assuming smooth marginal changes in recreational utility

B. Replacement cost approach based on historical engineering expenditures

C. Choice experiment incorporating state-dependent utility functions and ecosystem resilience metrics

D. Hedonic pricing focused strictly on immediate property value premiums

3. When analyzing hierarchical ecological data with spatial autocorrelation across multiple nested watersheds, which statistical framework provides the most robust parameter estimates while accounting for random intercepts and spatial covariance structures? [Week 12 | Hard]

A. Ordinary Least Squares (OLS) linear regression with standard error correction

B. Spatial autoregressive generalized linear mixed models (GLMMs) with explicit covariance matrices

C. Univariate ANOVA without spatial random effects

D. Standard principal component analysis (PCA) on raw environmental variables

4. During adaptive radiation driven by ecological character displacement, what primary mechanism prevents resource-mediated competitive exclusion between two sympatric morphs? [Week 11 | Medium]

A. Stabilizing selection acting uniformly on intermediate phenotypes

B. Directional divergence in resource utilization traits reducing niche overlap below the limiting similarity threshold

C. Random genetic drift operating independently of resource abundance

D. High rates of continuous gene flow homogenizing allele frequencies across populations

5. Urban fragmentation often creates matrix conditions that favor generalist species over habitat specialists. Which structural intervention best mitigates edge effects and facilitates wildlife movement across urban-wildland interfaces? [Week 10 | Easy]

A. Increasing the density of impervious paved surfaces to deter movement

B. Establishing continuous, structurally complex green corridors integrated with wildlife overpasses

C. Fragmenting remaining patches further into isolated pocket parks

D. Implementing intensive artificial lighting to illuminate migratory pathways

6. In ecological restoration, when a degraded system displays alternative stable states, what is the primary objective of applying a strategic disturbance regime prior to community re-assembly? [Week 9 | Medium]

A. To accelerate soil erosion and nutrient leaching to reset baseline conditions

B. To push the system across an ecological threshold, dismantling the undesirable stable state and enabling self-sustaining succession

C. To permanently sterilize the microbial community to prevent competition with planted cultivars

D. To eliminate all native seed banks and enforce monoculture plantation forestry

7. When designing a reserve network under climate change projections, which conservation planning strategy best addresses species range shifts while maintaining population viability? [Week 7 | Hard]

A. Designating static, isolated mega-reserves based strictly on historical distribution maps

B. Prioritizing climate-corridor connectivity between altitudinal and latitudinal gradients coupled with dynamic reserve portfolios

C. Focusing solely on endemic species richness hotspots without regard for future dispersal barriers

D. Relying exclusively on captive breeding programs without habitat preservation

8. Global climate change frequently induces phenological mismatches between interacting trophic levels, such as pollinators and flowering plants. What is a primary demographic consequence of this decoupling for specialized consumer populations? [Week 6 | Medium]

A. Exponential population growth due to increased resource availability

B. Decreased reproductive success and recruitment bottlenecks leading to population decline

C. Immediate phenotypic plasticity shifting the consumer's physiological thermal optima

D. Complete genetic purging of deleterious alleles within a single generation

9. According to island biogeography theory and landscape ecology principles, how does habitat fragmentation affect patch-level extinction and colonization dynamics? [Week 5 | Easy]

A. It decreases extinction rates and increases immigration rates across all taxa

B. It reduces patch area and increases isolation, simultaneously increasing extinction rates and decreasing colonization rates

C. It has no measurable impact on metapopulation persistence or gene flow

D. It transforms the surrounding matrix into a highly permeable and supportive habitat

10. In biogeochemical cycling, how does an acute shift from oxic to anoxic conditions in lake hypolimnions fundamentally alter internal phosphorus loading and subsequent primary productivity? [Week 4 | Hard]

A. Iron-bound phosphorus ($FePO_4$) is reduced, releasing soluble phosphate (PO_4^{3-}) into the water column and potentially triggering cyanobacterial blooms

B. Phosphorus is permanently precipitated into deep sediments as inert apatite minerals

C. Nitrification rates increase exponentially, consuming all available inorganic phosphorus

D. Microbial uptake of phosphorus ceases entirely, causing phosphate concentrations in the water column to drop to zero

Part B: Short Answer

Answer in the space provided. Show your reasoning clearly.

1. (8 pts) Consider an emerging ecological challenge involving the rapid proliferation of a novel viral pathogen transmitted across an expanding wildlife-livestock-human interface due to climate-driven range shifts. Propose an innovative, technology-driven surveillance and intervention framework that integrates eDNA metabarcoding and machine learning predictive modeling. Detail how this framework addresses both the epidemiological risk and the underlying ecological drivers, ensuring adaptive management under deep uncertainty. [Week 14 | Hard]

2. (8 pts) A municipal government is deciding whether to invest heavily in engineered grey infrastructure (e.g., concrete seawalls and water treatment plants) or nature-based green infrastructure (e.g., mangrove restoration and wetland expansion) to protect a coastal urban center from storm surges and nutrient runoff. Applying ecological economics principles, construct a comparative valuation framework. Contrast how the two options account for provisioning, regulating, and cultural ecosystem services, and explain how discounting future values impacts the economic viability of the green infrastructure investment. [Week 13 | Hard]

3. **(8 pts)** Researchers are investigating the drivers of plant species richness across a fragmented landscape containing 50 forest patches. The dataset exhibits severe spatial autocorrelation and zero-inflation in rare species counts. Outline an advanced statistical modeling strategy to analyze this data. Compare the utility of Generalized Linear Mixed Models (GLMMs) incorporating spatial covariance structures (e.g., conditional autoregressive priors) with Random Forest spatial interpolation, and explain how you would validate the model to avoid overfitting. [Week 12 | Hard]

4. **(8 pts)** In a high-elevation alpine system, a specialist plant species and its exclusive mammalian seed disperser are experiencing asymmetric range shifts driven by rapid warming. The plant is migrating upward faster than the disperser due to wind-dispersed pollen, while seed dispersal is becoming severely bottlenecked at lower elevations. Analyze the evolutionary consequences of this disruption. Discuss the potential for evolutionary rescue, the likelihood of shift from mutualism to parasitism or extinction debt, and how divergent selection pressures might alter floral and fruit traits over the next 50 generations. [Week 11 | Hard]

5. **(8 pts)** An expanding metropolitan area interfaces directly with a protected predator-rich forest, leading to escalating human-wildlife conflict involving apex carnivores (e.g., wolves and cougars) preying on domestic livestock and entering suburban neighborhoods. Propose an integrated urban ecology management plan based on the principles of landscape connectivity and behavioral ecology. Detail how you would utilize source-sink dynamics, ecological traps, and non-lethal deterrents (such as sensory habituation mitigation and corridor zoning) to achieve coexistence. [Week 10 | Hard]

Part C: Essay / Integrative

Answer in the space provided. Show your reasoning clearly.

1. **(20 pts)** Scenario: Consider a fragmented tropical forest landscape experiencing unprecedented microclimatic shifts and novel pathogen outbreaks driven by global climate change (Week 6). Conservationists propose deploying a geoengineered ecosystem intervention combined with assisted gene flow to save an endemic bird species.

   Task: Evaluate the ethical, evolutionary, and biogeochemical ramifications of implementing this emerging bio-intervention. Specifically, analyze how introducing genetically altered populations might disrupt local co-evolutionary dynamics (Week 11) and alter carbon sequestration pathways within the disturbed ecosystem (Week 4). Your response must synthesize principles of contemporary ecological risk assessment with advanced evolutionary theory. [Week 14 | Hard]

2. **(20 pts)** Scenario: A regional watershed authority is attempting to quantify and monetize the provisioning and regulating ecosystem services of an interconnected wetland and forest mosaic that has suffered severe habitat fragmentation (Week 5).

   Task: Critically evaluate the limitations of applying classical ecological economics and Total Economic Value (TEV) frameworks to highly fragmented and degraded landscapes. In your synthesis, integrate principles of spatial landscape ecology—specifically patch-corridor-matrix dynamics—to explain how structural connectivity influences the sustainable delivery and valuation of watershed services. [Week 13 | Hard]

HOW TO PREPARE — STUDY RECOMMENDATIONS

- Allocate your study hours proportionally, dedicating the majority of your time to the heavily weighted recent topics from Weeks 9 through 14.
- Construct conceptual synthesis maps that explicitly link early foundational concepts from Weeks 1 through 5 to advanced contemporary issues in Weeks 13 and 14.
- Practice interpreting complex graphical data sets and statistical outputs to prepare effectively for the advanced problem-solving required in Part B.
- Draft comprehensive essay responses focusing on how evolutionary ecology and ecological economics intersect with conservation and restoration practices.
- Form collaborative study groups to debate emerging challenges in ecology, ensuring you can articulate complex arguments at Bloom Level 5 (Synthesizing).

INSTRUCTOR COPY — ANSWER KEY & MODEL ANSWERS

Part A — MCQ Answer Key

Q	Answer	Q	Answer
1	**B**	6	**B**
2	**C**	7	**B**
3	**B**	8	**B**
4	**B**	9	**B**
5	**B**	10	**A**

Solution Explanations:

- **Q1 (B):** Option B is correct because dynamic probabilistic life-cycle assessment combined with PBTK-TD models allows researchers to simulate complex, time-varying exposures across food webs. Option A is a distractor because static models fail to capture the transient spikes and physiological feedback loops inherent to engineered nanomaterial toxicity.
- **Q2 (C):** Option C is correct because choice experiments with state-dependent utility can account for abrupt changes in ecosystem state and catastrophic welfare losses. Option A is incorrect because the travel cost method assumes smooth, continuous utility functions and fails near ecological tipping points.
- **Q3 (B):** Option B is correct as spatial GLMMs properly partition variance across hierarchical nesting levels while modeling spatial dependence explicitly. Option A is wrong because OLS violates the assumption of independence when spatial autocorrelation is present, leading to inflated type I error rates.
- **Q4 (B):** Option B is correct because character displacement minimizes interspecific or inter-morph competition by driving phenotypic divergence in resource-use traits. Option D is incorrect because high gene flow counteracts divergent selection, preventing local adaptation and speciation.
- **Q5 (B):** Option B is correct because green corridors and wildlife overpasses enhance functional connectivity and reduce edge permeability for sensitive species. Option A is a distractor because impervious surfaces exacerbate fragmentation and urban heat island effects.
- **Q6 (B):** Option B is correct because overcoming alternative stable states requires targeted interventions that alter feedback loops maintaining the degraded state. Option D is incorrect because restoration aims to promote biodiversity, not monoculture plantations.
- **Q7 (B):** Option B is correct because dynamic portfolios and altitudinal/latitudinal corridors allow species to track shifting climatic envelopes. Option A is a distractor because static reserves become ecologically mismatched as climates warm and species migrate beyond reserve boundaries.
- **Q8 (B):** Option B is correct because missing peak resource abundance reduces energy intake during critical reproductive windows, depressing recruitment. Option A is incorrect because mismatch reduces resource availability, lowering fitness rather than increasing it.
- **Q9 (B):** Option B is correct because smaller patches support smaller populations prone to demographic stochasticity (higher extinction), while greater isolation reduces migrant rescue effects (lower colonization). Option A incorrectly reverses both core mechanisms.
- **Q10 (A):** Option A is correct because under anoxic conditions, ferric iron (Fe^{3+}) is reduced to ferrous iron (Fe^{2+}), dissolving iron-phosphate complexes and releasing bioavailable phosphorus that drives eutrophication. Option B is incorrect because anoxia promotes release rather than permanent burial of phosphorus.

Part B — Model Answers

Q1 (8 pts): Consider an emerging ecological challenge involving the rapid proliferation of a novel viral pathogen transmitted across an expanding wildlife-livestock-human interface due to climate-driven range shifts. Propose an innovative, technology-driven surveillance and intervention framework that integrates eDNA metabarcoding and machine learning predictive modeling. Detail how this framework addresses both the epidemiological risk and the underlying ecological drivers, ensuring adaptive management under deep uncertainty.

Model Answer / Key Points: An excellent response must include: (1) Integration of eDNA metabarcoding from water or soil samples to detect pathogen presence and host species distributions non-invasively before clinical outbreaks occur; (2) Implementation of machine learning algorithms (e.g., random forests or neural networks) fed with climate, spatial fragmentation, and host density variables to forecast high-risk transmission hotspots; (3) A proactive adaptive management loop where model predictions dynamically direct targeted vaccination or movement restrictions; (4) Consideration of socio-ecological feedbacks, recognizing that human land-use changes amplify contact rates, thereby linking conservation actions with public health outcomes.

Q2 (8 pts): A municipal government is deciding whether to invest heavily in engineered grey infrastructure (e.g., concrete seawalls and water treatment plants) or nature-based green infrastructure (e.g., mangrove restoration and wetland expansion) to protect a coastal urban center from storm surges and nutrient runoff. Applying ecological economics principles, construct a comparative valuation framework. Contrast how the two options account for provisioning, regulating, and cultural ecosystem services, and explain how discounting future values impacts the economic viability of the green infrastructure investment.

Model Answer / Key Points: An excellent response must include: (1) Identification of specific ecosystem services: regulating services (storm surge attenuation, nutrient sequestration), provisioning services (fisheries support), and cultural services (recreation, aesthetic value); (2) Use of valuation methodologies such as replacement cost, contingent valuation, or avoided damage cost for the nature-based solution; (3) Explicit analysis of dynamic ecosystem service generation over time, noting that green infrastructure appreciates in value as ecosystems mature, unlike grey infrastructure which depreciates; (4) Discussion of discount rates, explaining how high discount rates unfairly penalize long-term benefits of green infrastructure, and advocating for hyperbolic or lower social discount rates for sustainable environmental assets.

Q3 (8 pts): Researchers are investigating the drivers of plant species richness across a fragmented landscape containing 50 forest patches. The dataset exhibits severe spatial autocorrelation and zero-inflation in rare species counts. Outline an advanced statistical modeling strategy to analyze this data. Compare the utility of Generalized Linear Mixed Models (GLMMs) incorporating spatial covariance structures (e.g., conditional autoregressive priors) with Random Forest spatial interpolation, and explain how you would validate the model to avoid overfitting.

Model Answer / Key Points: An excellent response must include: (1) Addressing zero-inflation by proposing a Zero-Inflated Poisson (ZIP) or Zero-Inflated Negative Binomial (ZINB) framework to handle excess zeros; (2) Incorporating spatial autocorrelation by adding a spatial random effect via GLMMs with spatial covariance matrices (e.g., CAR/SAR) or Spatial Eigenvector Mapping (SEVM); (3) Contrasting GLMMs (parametric, provides explicit parameter estimates and ecological inference) with Random Forest spatial approaches (non-parametric, excellent at capturing complex non-linear interactions but weaker for direct mechanistic inference); (4) Describing spatial cross-validation (e.g., leaving out spatial blocks rather than random points) to prevent spatial leakage and reliably test out-of-sample predictive accuracy.

Q4 (8 pts): In a high-elevation alpine system, a specialist plant species and its exclusive mammalian seed disperser are experiencing asymmetric range shifts driven by rapid warming. The plant is migrating upward faster than the disperser due to wind-dispersed pollen, while seed dispersal is becoming severely bottlenecked at lower elevations. Analyze the evolutionary consequences of this disruption. Discuss the potential for evolutionary rescue, the likelihood of shift from mutualism to parasitism or extinction debt, and how divergent selection pressures might alter floral and fruit traits over the next 50 generations.

Model Answer / Key Points: An excellent response must include: (1) Analysis of the breakdown in co-evolutionary selection pressures, leading to a geographic mosaic of co-evolution where mutualistic selection is relaxed at the leading edge; (2) Evaluation of evolutionary rescue potential, noting that genetic variation in

the plant must be sufficient to adapt to dispersal failure (e.g., shifting toward selfing or abiotic dispersal) within the timeframe of environmental change; (3) Discussion of extinction debt, where surviving populations persist transiently despite being ecologically doomed by demographic collapse; (4) Prediction of trait evolution: directional selection favoring reduced investment in costly fleshy rewards (fruit traits) and increased autonomous self-pollination or wind-dispersal mechanisms if the disperser is permanently lost.

Q5 (8 pts): An expanding metropolitan area interfaces directly with a protected predator-rich forest, leading to escalating human-wildlife conflict involving apex carnivores (e.g., wolves and cougars) preying on domestic livestock and entering suburban neighborhoods. Propose an integrated urban ecology management plan based on the principles of landscape connectivity and behavioral ecology. Detail how you would utilize source-sink dynamics, ecological traps, and non-lethal deterrents (such as sensory habituation mitigation and corridor zoning) to achieve coexistence.

Model Answer / Key Points: An excellent response must include: (1) Application of source-sink theory, identifying suburban residential zones as ecological traps where abundant food attracts carnivores but high mortality risk turns them into population sinks; (2) Design of functional wildlife corridors using landscape connectivity principles to guide movement away from dense human settlements; (3) Implementation of non-lethal deterrents combined with variable reinforcement schedules to prevent sensory habituation; (4) Addressing socio-ecological dimensions, such as securing anthropogenic food attractants (garbage, livestock) to eliminate the positive reinforcement driving urban foraging behavior.

Part C — Model Answers

Q1 (20 pts): Scenario: Consider a fragmented tropical forest landscape experiencing unprecedented microclimatic shifts and novel pathogen outbreaks driven by global climate change (Week 6). Conservationists propose deploying a geoengineered ecosystem intervention combined with assisted gene flow to save an endemic bird species.
Task: Evaluate the ethical, evolutionary, and biogeochemical ramifications of implementing this emerging bio-intervention. Specifically, analyze how introducing genetically altered populations might disrupt local co-evolutionary dynamics (Week 11) and alter carbon sequestration pathways within the disturbed ecosystem (Week 4). Your response must synthesize principles of contemporary ecological risk assessment with advanced evolutionary theory.

Model Answer / Key Points: An excellent response must cover the following points: 1. **Evolutionary Ramifications**: Analyze how assisted gene flow and genetically modified traits interact with local selective pressures, potentially leading to maladaptation or outbreeding depression under rapid climate change. 2. **Biogeochemical Feedback**: Discuss how altered consumer-resource dynamics resulting from population replacement impact local nutrient cycling rates and carbon sequestration (e.g., changes in herbivory altering litter quality and decomposition rates). 3. **Risk Assessment**: Evaluate the unintended consequences of geoengineering and synthetic biology in wild ecosystems, drawing upon frameworks from contemporary ecology (Week 14) and evolutionary trajectories (Week 11).

Q2 (20 pts): Scenario: A regional watershed authority is attempting to quantify and monetize the provisioning and regulating ecosystem services of an interconnected wetland and forest mosaic that has suffered severe habitat fragmentation (Week 5).
Task: Critically evaluate the limitations of applying classical ecological economics and Total Economic Value (TEV) frameworks to highly fragmented and degraded landscapes. In your synthesis, integrate principles of spatial landscape ecology—specifically patch-corridor-matrix dynamics—to explain how structural connectivity influences the sustainable delivery and valuation of watershed services.

Model Answer / Key Points: An excellent response must cover the following points: 1. **Economic Valuation Limits**: Explain the shortcomings of monetizing ecosystem services when ecological functions are non-linear, threshold-dependent, and heavily compromised by fragmentation. 2. **Landscape Connectivity Integration**: Apply patch-corridor-matrix models to demonstrate how physical fragmentation disrupts hydrological flows and nutrient retention, thereby eroding regulating services. 3. **Synthesis**: Propose a modified valuation framework

that incorporates spatial metrics from landscape ecology (Week 5) to account for edge effects and matrix permeability when calculating economic returns on ecological restoration.